Exercise 1

Description of audio processes in Audacity.

Recording:

To start recording we need to click Record button (red circle).
To stop recording we need to click Stop button (black square).
Audacity also allows us to choose recording device. Recording automatically creates new track where we can see spectrum of our recording.

Editing:

By selecting track, or track part we can perform various editing tasks. We can see full list by clicking edit button.
For example, delete feature allows us to delete parts of silence we do not need in this audio exercise.

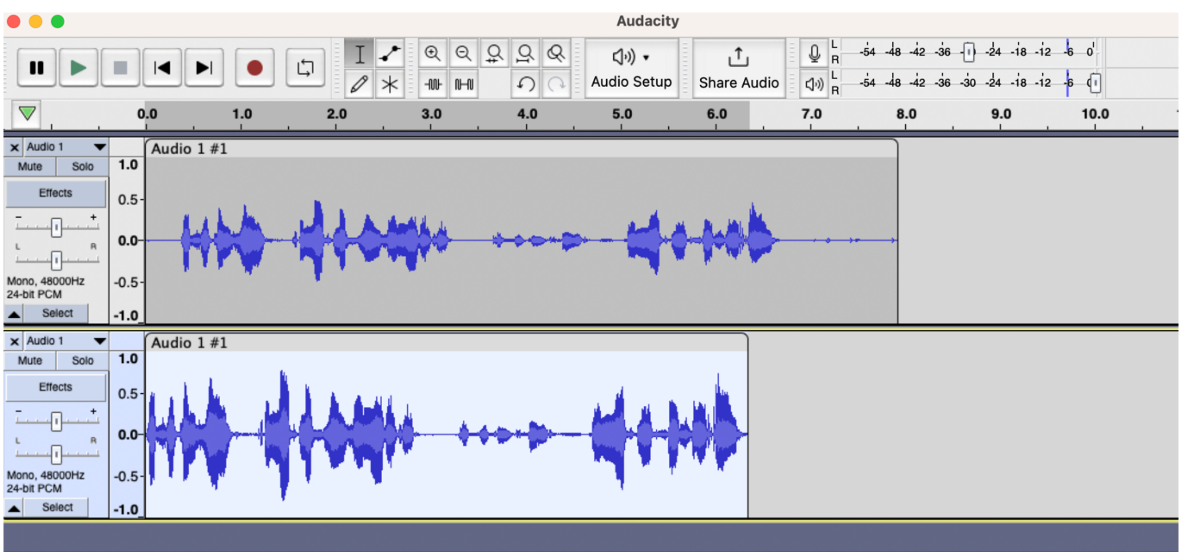
Processing:

We can select audio and process it through various audio effects. Audacity allows us to choose the track or part of track we want to apply effect to, and then by clicking Effect we can choose one of build in effects.

Saving:

Saving is done by exporting audio file, Audacity allows us to set up export parameters, type of file etc.

Photo from editing audio file – delete silence, normalize audio



Description of effects – and how they have been programmed.

Low-pass filter:
description: cuts out audio that is higher that set frequency.
programmed: by creating new p5.LowPass object in setup function, then I created function that applies changes to that object when according sliders are changing

Waveshaper Distortion:
description: allows to change the shape of waves from more sinusoidal to more rigid etc.
programmed: by creating new p5.Distortion object in setup function, then I created function that applies changes to that object when according sliders are changing

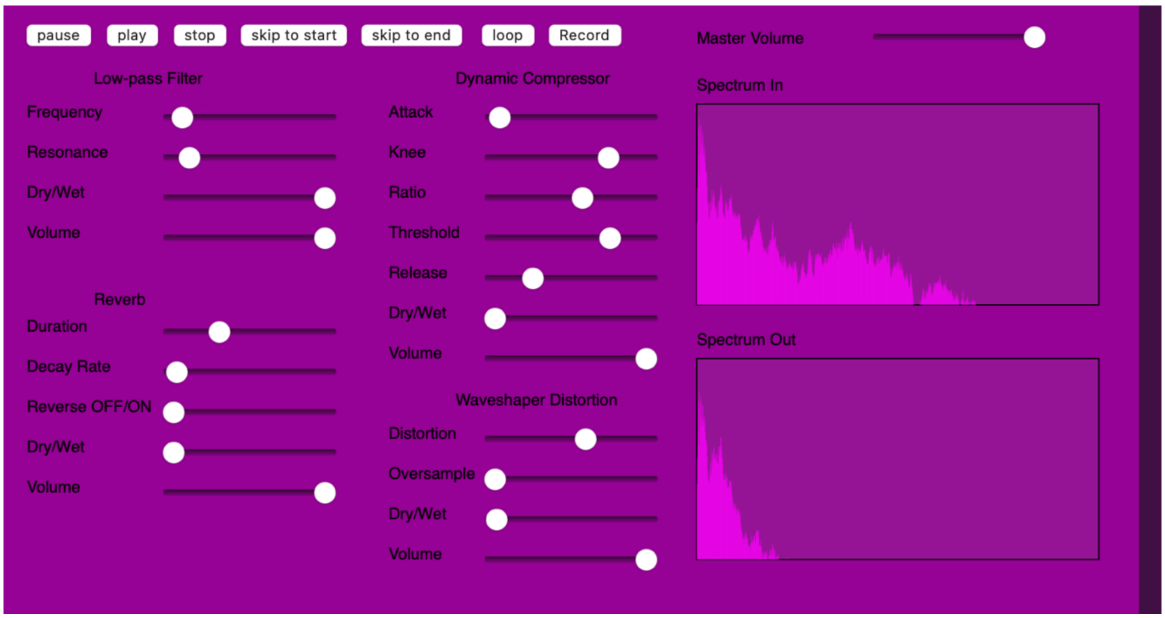
Dynamic Compressor:
description: allows to change the volume of sound sample dynamically meaning we can set the level of volume we desire, and compressor dynamically will adjust amplification for that frame to match desired amount, making quite sound louder and loud sounds quieter
programmed: by creating new p5.Compressor object in setup function, then I created function that applies changes to that object when according sliders are changing

Reverb:
description: it allows to make effect that audio has been recorded at certain surrounding how it echoes etc.
programmed: by creating new p5.Reverb object in setup function, then I created function that applies changes to that object when according sliders are changing

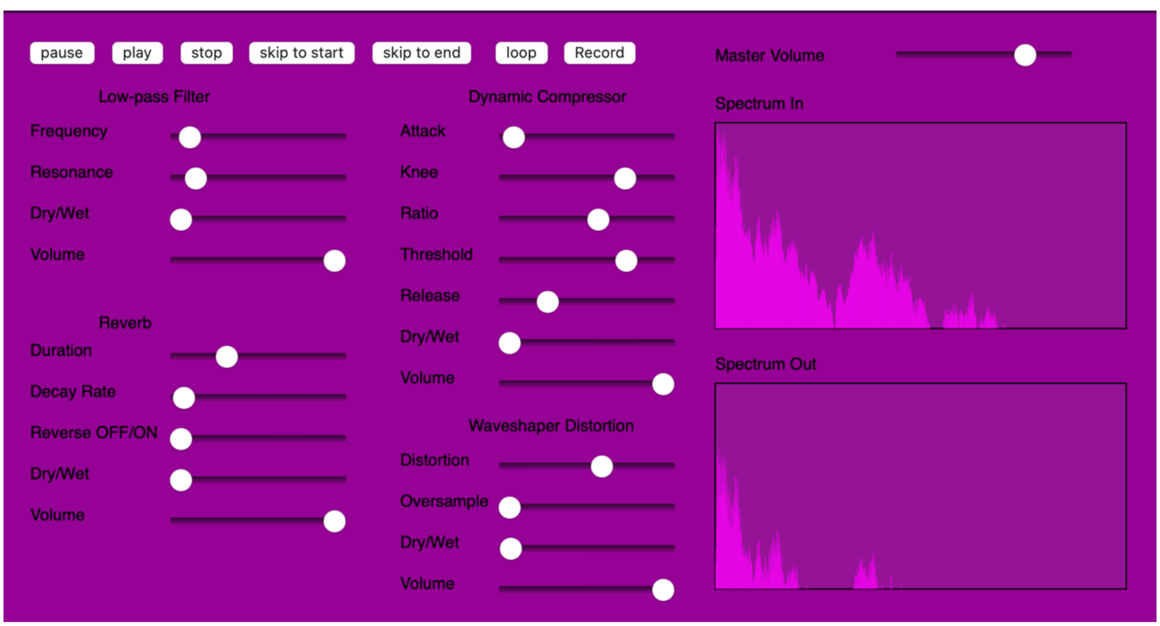
Master Volume:
description: allows to change volume of the filtered file
programmed: to make sure that master volume is changing the amplitude value as the last step of audio processing chain I used outputVolume function that changes volume of output without effecting volume of input file.

Analysis of application – low-pass filter and master volume effecting sound spectrum

Low-pass filter – spectrogram shows amplitude over frequency by applying low pass filter we can see that after cut of frequency amplitude is 0 for higher frequency. Meaning the filter worked sound was only passed for low frequencies.



Master Volume - it effects spectrogram by lowering amplitude by certain value for every frequency evenly. We can see that in the photo from application below. Shape of the spectrum waves is unchanged but the heighted of values on the spectrogram is significantly lower.



Link to lab:
https://hub.labs.coursera.org:443/connect/sharedtfolfbtj?forceRefresh=false&path=%2FvMBxTlyvUcM5DXb37Upsye6EhaJPW4z7oTLHiHdIrBJcLsAIkbpg8LFmKA6qdMB0%2F&isLabVersioning=true